

ated by speakers on either side of the pool of fuel. In this method, fire was extinguished using two different effects. One, sound waves increased the air velocity, which thinned the flame boundary area, making it easier to dislodge the flame.

Second, the acoustic field disturbed the pool surface, causing the flame to widen and decrease in temperature. Essentially, the flame extinguished as the same amount of heat spread over a larger area, with speakers blasting sounds at specific frequencies that extinguished the flame.

Electric fire extinguisher

In Indian mythology, people bestowed with super-natural powers used to create fires out of nothing and extinguish them by waving their hands. These powers might be based on the technique of generating electric fields to a level that these can control fires.

In this present era, life is based on the use of electricity—electricity is almost everywhere and it is appropriate to have a device based on electricity to control fire. **An electrical fire extinguisher is basically a thin wire that generates oscillating electric fields, which can put out the fire when directed at the flame.**

According to scientists, the first report on the electrical nature of flame dates back to 200 years. They observed that electricity could affect the shape of flame, making it bend, twist, turn, flicker and even snuff it out. Controlling fire is an enormously difficult challenge. Although, scientists have known for years that electricity can be used to fight fires, all previous attempts have been based on the use of direct current (DC)-producing static electric fields, and effects were too small to be practical.

Recently, researchers have discovered that application of alternating current (AC)-producing oscillating electric fields can prove to be more beneficial, and that application of large electric fields can be used to suppress flames rapidly.

Harvard University's chemist Ludovico Cademartiri has shown, at 2011 American Chemical Society (ACS) conference, that using AC (oscillating electric fields) can multi-

ply the effects of the electric field. By applying an oscillating electric field, stronger interactions between the electrical field and charged flame particles could be triggered, and effects were strong enough to put the flame out.

Although the working phenomenon is complex with several effects occurring simultaneously, among these effects, it appears that carbon particles (or soot) generated in the flame are the main components for its response to an electric field. As soot particles can easily become charged, these charged particles respond to the electric field, thus affecting stability of the flame. When attached to an amplifier, the wand can snuff out the flame, and its effect increases as voltage increases.

To put out the flame, the approach is to get very close to it (right in front) and point the wire at the flame. **The concentrated electric field that emerges out of the tip of the wire should be pressed on the flame until it (flame) separates from its fuel source.** The electric field, which has the strength of a million volts per metre, exerts a force on charged particles in the flame. Positive and negative particles accelerate in opposite directions. When flow of the charge is high enough, the flame is pushed off the source.

Developmental status

In the new study, scientists connected a powerful electrical amplifier to a wand-like probe and used the device to shoot beams of electricity at an open flame more than 30.5cm (1-foot) high. Almost instantly, the flame was snuffed out, and it worked time and again. The device was made to operate with a 600-watt amplifier, which is almost of the same power as a high-end car stereo system.

However, they believe that a power source with only a tenth of this wattage could have similar flame-suppressing effects. At present, the technology looks great in the laboratory, but translating it to real fire-fighting scenarios could prove to be difficult. In the immediate future, perhaps the system can be more useful in places where fires start out small.

Its comparative efficacy has not been tested, but electricity is likely to be used as an alternative to, rather than

a replacement for, traditional firefighting equipment. Firefighters could theoretically clear a path through a larger fire using the electrical power wand, or put out a small fire without causing water or foam damage to property.

Use of electric fire extinguishing technology can also save water and avoid the use of other environmentally harmful fire-fighting materials.

Scientists believe that the technology would work best for fires in such confined spaces as aboard submarines, but not so much in wide open areas like forests. They speculate that the technology could be built into a vehicle or a ship hold's fire-suppression system.

This new technology could also put out or suppress fires remotely.

The future device

Scientists envision that futuristic electric fire extinguishers could be fixed on the ceilings of buildings, airplanes, ships or vehicles, just like stationary water sprinklers or carbon-dioxide fire extinguishers that are currently used.

Alternatively, firefighters could carry electrical wave blasters in the form of a backpack and use electricity to control fires using a handheld wand. Such type of devices could also be used to make a path for firefighters to enter a fire or create an escape path for trapped people to get out.

Further, if an high-intensity AC electric field can influence and attract oppositely-charged carbon particles in a fire—snuffing out small fires but not enough for large fires—then maybe an atomised fine spray of oppositely-charged water could attract oppositely-charged fine particles produced in huge forest fires.

An apparatus that atomises water is needed in combination with electrodes close to output atomised water. Electrodes would be connected to a powerful AC electric field device. This might work even better than the electric field alone for large forest fires.

Fine water droplets may be attracted more efficiently to carbon particles with little wastage of water, possibly smothering the fire rapidly by denying fuel for further combustion. However, a lot of research is still needed to figure out how long a fine water droplet could hold an opposite charge. **EFY**